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Chapter 21
A Case Study: Using Game Theory

Over the course of Part III, we have spent a lot of time looking at various toy and real games, attempting to obtain insight into how the various dimensions of the game will affect optimal strategy. One of the hardest things to do, however, is to put these largely theoretical concepts into practice at the table. We stated at the outset of this book that we would avoid writing a manual on actual play, and even in this chapter, we will speak largely in generalities (particularly because of the complexity of factors that affect our decisions at each point). However, it seems that the best way to illustrate the process of applying theory in practice is to provide an overview and some examples of the way that ***we*** play particular situations in a particular game, demonstrating ways in which some of the various results we have obtained through the course of this book can be applied in practice.

Of course, we are not claiming that our strategy is optimal, unexploitable, or perfectly balanced. In fact, we know that this is not the case, because we know how difficult it is to play in an entirely balanced manner even in the simplest games. But hopefully you will see how our decisions are primarily influenced by two concepts: playing with balance and having an appropriate level of aggression for our value actions. The game we have chosen is the poker game that is most popular at this writing: no-limit holdem.

In discussing any no-limit game, the stack sizes are of paramount importance. In certain parts of this discussion, we will discuss situations and ideas that cover different stack sizes, while in others, we will limit the discussion to a specific stack size. In our discussion here, we will ignore our opponents' tendencies or tells as a source of information. Because poker is a multiplayer game, there are no optimal strategies (more on this in Part V). But we are searching for strategies that are near-optimal, or at least balanced, which can be profitably played against a wide range of opposition with little or no information.

A thorough discussion of this nature of a game as complex as no-limit holdem could likely fill more than one volume the size of this one; so we will try to keep our comments brief. You will also notice that we do not discuss "hands" here; a particular hand is of very little significance to us. Playing a particular betting sequence (including the hand distribution you carry to that sequence: is. what near-optimal play is all about, and so we describe situations without reference to the cards we hold.

We begin by discussing preflop play (when we are the first player in).

Preflop Play (When First In)

The problem of selecting a strategy for play preflop is not an easy one. There are a number of factors that we must consider when choosing the hands with which we will enter the pot, and the strategy we will employ when doing this. Consider the case where we are first in; that is, all the players in front of us have folded

A preflop strategy is essentially a probability distribution for each hand, including how often we will raise (and to how much) with a hand, how often we will call with a hand, and how often we will fold the hand. Our experience examining games such as the [0,1] game indicates that many of these decisions will be pure; that is, most hands will not utilize mixed strategies. In general, hands that have mixed strategies will be threshold hands. In holdem, however, , card removal effects are of some significant concern, and so we expect to find that more hands have mixed strategies than only those hands at the thresholds.

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Identifying this strategy is no simple task, especially because even though we can break the strategy down into different elements, these elements are strongly tied together. For example, we can subdivide our preflop strategy into three major areas.

- Which hands to play
- Whether to raise or call with those hands
- What amount to raise with hands that will raise

Each of these items is somewhat complex in it's own right: but they are also directly tied together. For example, suppose we are considering whether we should play 88 from six off the button with 20 big blinds in our stack. Our overall strategy for playing hands from this position will have a strong influence on our decision to play or not play this hand. Suppose we make a strategic decision to limp with all hands with which we will continue. Then 88 would seem to have positive equity when played with such a strategy. However, if our strategy is to raise to four times the blind from this position with all hands we will play. 88 is likely a money loser and should be folded.

And yet, in addressing the balance of the strategic alternatives we choose to employ across our entire strategy, we need information about each particular hand. Suppose for a moment that we decide to raise first in with any hand with which we will enter the pot, and to fold other hands. Then the only task that remains is to select the hands which merit a raise based on their expectation as part of a distribution of hands that plays in this manner. We might decide to limp in with any hand we will play preflop, folding the rest. Then all we need decide is whether to play a hand by calling or not, a decision that rests on the expectation of each of our hands when played as part of a distribution of hands that limps in this situation all the time. A third possibility is that we choose to employ a strategy of limping sometimes and raising sometimes. If we choose this alternative, then we are forced to consider the balance of both our raising and calling hands to ensure that our strategy does not give away enough information that it can be exploited. The point here is to consider the value of our entire distribution and find the distribution that maximizes our total expected value per hand.

This circularity makes defining a near-optimal preflop strategy extremely difficult. What we do in practice, however, is to choose to hold one of these variables constant, and from that make inroads to the others. It is our preference to play a strategy that never calls first in outside the blinds. There are a number of reasons for this. First, from each position, we will be playing a distribution of hands that is strong relative to the random hands left behind us. As a result, we expect to have the best hand, and benefit from more money going into the pot preflop. Second, we believe that the value of putting pressure on the big blind to call a raise in order to see the flop is higher than the potential lost value from raising instead of calling when our opponents hold strong hands. As we saw throughout Part III, making the opponent indifferent is an important principle of optimal play; in a sense we are attempting to create some land of indifference for the blind in cases where it is folded around to him. Third, we generally feel that the potential loss from the fact that we can play fewer hands hi a raise-only strategy than we could in a strategy that was mixed between calling and raising is more than offset by the information hiding that playing a uniform strategy affords us.

The next major concern is that of position. Play from the front is perhaps the most poignant example of the power of position. Six other (presumed near-optimal) players will act after us, and each has the possibility of picking up a strong hand. Entering the pot with weak or marginal hands will likely result in being raised or reraised when one of those players has a strong hand. Therefore, we must take care in the hands that we select to play.

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One of the principles of play in the front is to be selective, because of the threat of all the hands behind us.

As we move around toward the button, there are fewer and fewer random hands behind us, and so we can raise with weaker and weaker hands. On the button, then, we can play some wide variety of hands that rate to have positive expectation against the blinds, including the value of position on later streets. Still, though, we must take care to play only loosely enough to add hands that enhance the value of our distribution.

To find these hands, we consider our distribution as containing zero hands. Then we can add hands that have positive expectation one by one, until we can no longer add hands with positive expectation to our distribution. For each hand, however, we consider the value of the hand in the context of the new distribution. To understand what we mean by this, consider a simple example. Suppose that the first hand we add is AA, the strongest hand. Just assume temporarily that we are choosing to raise all hands first in, and that we raise to double the big blind in first position with AA and no other hands.

In our quest to be unexploitable, we can even take the position that we are going to announce to the table our strategy beforehand. Suppose we did this and our strategy was to raise AA alone. Then clearly our opponents would never reraise our raises, and would likely fold all their hands, provided the stacks were not terribly deep. So our equity here with AA would be something approximating the blinds. Obviously we can do better; suppose we add KK to the mixture. Our distribution is still quite strong, but the blind can probably call with hands containing a single ace. Still, the equity of each raise we make with a distribution of {KK+} is going to be worth approximately the blinds.

Suppose we go on adding hands to our distribution: QQ, AKs, Ako, and so on. At some point, we will have a weak enough distribution of hands such that our opponents will want to play with us especially out of the blind, but even in the field. When this occurs, it should be evident that though the value of the hands on the borderline is relatively small, the value of AA has increased. So when we raise with a distribution like {99+. AQ+], AA has much higher value than when we raised it out of a distribution of {KK+}. In poker, it is generally very difficult to reduce the value of the pre-existing hands in a distribution by adding new, weaker hands. In fact, the pre-existing hands normally gain value by doing this. Constructing a distribution that maximizes the total value of our hands should be our goal.

In doing this, we must also take into account card removal effects and the effects of the community cards on the future play. In most of the games we have studied so far (such as the [0,1] game), hand values have been monotonic; that is, a particular hand is either stronger or weaker than another. However, the nature of the community cards in holdem weakens this assumption. Take, for example, the two hands AJs and 98s. Against a random hand (and against most normal distributions of hands), AJs is stronger. However, a difficulty arises because the flop has such a significant impact on hand values. There are many flops where the threat of holding 98s will add value to the play of the entire distribution of hands. This holds for many different types of hands. We saw in the chapters on multi-street play (Chapters 19 and 20) how important it is to threaten to have strong hands. Particularly when the stacks are deep, this effect becomes more and more pronounced. As the stacks deepen, we add more and more hands (particularly in fractional increments, using mixed strategies) to our raising distributions. The types of hands we add are hands which threaten to have or make the nuts on typical types of flops, including weaker pairs, AXs, suited connectors, and so on.

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But this generally does not extend to playing weak hands for "deception" purposes. Optimal or near-optimal play does not contain such loss-leader plays. The ability to threaten to hold the nuts in a particular situation is worth some equity to all the hands in our distribution - but playing a hand with inferior showdown potential costs equity as well. With gigantic stacks (say, one million times the blind), it would indeed be indicated to play 52s under the gun with some positive probability, in order to threaten to have the nuts on boards with A34 of that suit. However, it would not be indicated to ever play 52o from under the gun.

The final piece of our first-in preflop discussion is to consider raise amounts. To understand the logic that we use to define the amounts we will raise, we begin by considering the conflict between the raiser and the blind. The blind is a key player in every hand. This is because the blind is the player most likely to call a raise from the field, due to the favorable odds afforded him by the dead money in the pot. There are two primary situations that face the raiser. When our opponents have a strong hand in the field, they will likely reraise or at least call. When no one behind us has a strong hand, it will be typical that the pot will be contested headsup between the raiser and the blind. In choosing a raise amount, we will want to strike a balance between these two scenarios. In the first situation (where an opponent has a strong hand), we generally want to keep our investment in the pot small. By doing this, we lose less on the hands we have to fold to his reraise, while still retaining the option to commit more chips when we have stronger hands. Conversely, in the second scenario, we want to make our raise amount appropriate to put some pressure on the big blind while extracting the appropriate value. Raising to (for example) ten times the big blind with a strong distribution would force him to fold many hands where he would have enough equity to call a smaller raise, while if we were permitted by the rules to raise to 1,05 units, he would have an easy call with any two cards. We choose raise amounts that are intended to give the big blind a difficult decision when the field folds to our raise when he holds a hand of only moderate strength. We cannot make him indifferent to calling with his better hands - rather, we could do this, but it would sacrifice too much equity overall.

This means that our raise sizes should be varied according to the strength of our distribution, and as our position becomes later and later, we open with weaker and weaker hands. This is because the threat of players behind us is diminished as the number of potential hands behind us lessens. As a result, we can open with a weaker distribution of hands. To achieve the effect we just described when opening on the button with our weaker distribution, we have to raise a larger amount, So in our raise-only strategy, we choose a set of raise amounts that varies on position. We choose not to vary our raise amounts based on the strength of our hand -hopefully you realize that the information hiding implications of raising different amounts with different hands are indeed troublesome. The approximate amounts that we raise into a pot that contains only the blinds and no antes steadily increase from the minimum raise of doubling the big blind from positions five off the button or more to tripling the blind on the button. Antes generally cause us to raise a little more in order to attempt to preserve the same ratios for the blind. For example, suppose we would raise to 1200 to win 600 (200-400 blinds) from the button. Then the blind would be getting 1800 to 800 to call. Adding antes of 25 (eight players in), so that the pot is now 800, would make a similar ratio (if we raised to x) of $(x+ 800)$ to $(x- 400)$. This occurs where x — 1350. So the antes of 200 result in a larger raise amount of about 150.

To recap, then, we employ a preflop strategy that has the following elements.

- When first in, we raise with any hand with which we intend to enter the pot.
- Our raise amounts vary from the minimum of doubling the blind (when five or six off the button) to approximately three times the blind on the button. When there are antes present, we modify our raise amounts to keep the same ratios of call to pot size.

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- We play distributions of hands such that all the hands we play have positive expectation (as far as we can estimate) against the remaining field of near-optimal non-colluding players.

Jamming First In (Preflop Play with Short Stacks)

When we discussed raise amounts in the previous section, we assumed that the stacks were of sufficient size that we would always want to raise to some fraction of our stack and anticipate later play. However, as the size of our stack relative to the blind shrinks, we find ourselves in a situation where we may not want to raise to the smaller amounts described previously. Suppose that we have a stack size of just three big blinds and we are five off the button. We decide to raise with some hand. According to the guidelines of the previous section, we would raise to just double the big blind.

However, this would leave us in a very silly-looking position were the blind to call our raise. The pot would be 4 $^1/_2$ big blinds, and we would have remaining a stack of just one big blind. We would be hard-pressed to fold post-flop in such a situation. As a result, our opponent can effectively force us to commit at his option, either preflop or on the flop. Contrast this with the situation that would occur if we raised all-in preflop. Then our opponent lacks options: he can play or fold. Recall from Chapter 12 that strategic options have non-negative value. So, we should seek to prevent our opponent from having a strategic option at no cost. In this case, raising all-in is indicated in lieu of raising to two-thirds of our stack.

But suppose our stack is four, or six, or fifteen big blinds? At some point, our situation changes, and raising to the smaller amount is again indicated. This threshold is scarcely static - it clearly changes with position. But we can attempt to estimate where this threshold is in order to play more accurately preflop. We will take as our example a raise from early middle position (four off the button) and work backward from there. Suppose we are playing a distribution of hands (near the cutoff between jamming and raising a small amount) that is as follows:

{77+, AJs+, AQ+, KQs)

We can calculate the equity of this distribution against a typical reraising distribution, such as {TT+. AQs+, AK+}. Our raising distribution has showdown equity against this distribution of 41.9%. Using this number, it would seem that we would be committed to call a reraise as long as we had pot odds to call with this equity.

But this is a mistake - we are not interested here in how our entire distribution does against the reraiser. We know we are going to call an all-in reraise with aces! We are interested in the equity of our worst hands. In this case, our worst hands of each type are 77, which has 33.0% equity, AJs, which has 32.0% equity against the reraiser, and KQs, which has 31.88%.

Suppose that we identified a threshold *x*, which is the stack size such that if we raised the big blind of one to two units and were reraised all-in, we would be compelled to call. If we have a stack lower than *x*, then we remove a strategic option (flat calling the raise) from our opponent by jamming instead of raising to a small amount and therefore pick up whatever value that strategic option would have afforded him.

If we raise to 2, and someone behind us in the field raises to *x*, then we are calling *x-2* units to win *x* + 3.5 units. In order for it to be correct to call this bet with all hands, our equity in the resultant pot with our weakest hand must be greater than the cost of calling.

$(0.3188)(2x+ 1.5) > x - 2$

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$x > 6.83$

So for this example, we find that a direct lower bound for x is a little less than seven big blinds. Of course, this is only applicable to this particular distribution of raising hands and this particular reraising distribution. We can treat this problem a little more generally. Consider the following game theory ideas:

- We should be indifferent to jamming with the weakest hand in our distribution. This is because our choice is between folding and jamming. Since this hand is the threshold between folding and jamming, the EV of jamming should be approximately zero.
- At the stack size that is the threshold between jamming and raising to a smaller amount, we will be indifferent between these two options (although perhaps we will play different hands).

Suppose we have a near-optimal opponent behind us. (Ignore for a moment hands left to act after him.) If he knows that this is the stack size at which we are indifferent between jamming and raising a smaller amount, he should be willing to commit with the same distribution of hands when we raise a smaller amount as when we jam. If he is not, we may exploit him by jamming with particular hands in our distribution and raising to a smaller amount with others.

Use **N** to designate the distribution of hands with which he will commit all-in against our raise to x units. Then **N** contains all hands that have positive equity confronting our jam. Suppose we jam a stack of 6 units with the aforementioned range: (77+, AJs+, KQs). Then N consists of all hands that have at least $^{12}/_{27}$ of the pot, or 44.44%. This distribution is: (99+, AKs, AKo}. Against this range, our worst hand is 77, which has just 29.91% of the equity in the pot. However, had we raised to a smaller amount here, we would still be committed to calling an all-in reraise, as we have odds:

$0.2991(13.5) – 4 > 0$

Suppose we increased the stack size now to 8 units. Then the equity threshold for reraising rises to 16/35 of the pot, or 45.7%. This excludes 99 from the reraising range, leaving us with (TT+. AKs, AKo). Against this distribution, 77 is still the worst, with 31.41% of the pot. If we raised to two units, a call is no longer indicated:

$(0.3141)(17.5) – 6 <0$

Again, we see that somewhere between six and eight units is an appropriate cutoff if our only concern is whether we will be forced to call a reraise when we are in early position. We can perform the same calculation on a much wider raising range, such as when we are on the button:

{22+, A2+, K2s+, K8o+, Q8s+, Q9+, J8s+,JT, T8s+, 97s+, 87s)

Suppose here that we raise to 3 units out of our stack of 10 units, with the intention of calling any raises. Then N contains all hands that have positive equity against our distribution for 10 units. Hands in N have at least $^{20}/_{43}$, or 46.5% of the pot; this distribution is:

{33+, A2s+, A5o+, KTs+, KTo+, QJs}

This is a much wider distribution! After a reraise from this distribution, we would be calling 7 units to win a pot of 14.5. Our worst hand equity wise here is K8o, which has 33.4% equity against the reraising distribution. We can see that

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(0.334)(21.5) - 7 > 0

So at this stack size, it is indicated to call with all hands after a reraise at a stack size of 10. In addition to this, there are other potential benefits to pushing all-in; we must compare the equity of pushing all-in as a strategy to the equity of raising to a smaller amount (assuming we do not want to break information biding by raising two separate amounts).

There are a number of difficulties associated with this process. One is that it is fairly difficult to accurately estimate equity against a distribution when there is postflop play. A second difficulty is that many of the hands that have sufficient showdown equity to make them playable are very difficult to play after the flop. Consider the most basic example of this, AX offsuit. Suppose that we raise to a smaller amount, as above, such as 3 units from our stack of 12 units. On the flop, the pot will be 7.5 units and we will have 9 units left in our stack, a little more than a single pot-sized bet.

This stack size leaves little room to maneuver; after the opponent checks we have a difficulty. Obviously if an ace has hit the flop we are willing to commit all our chips; the same is true if we paired our kicker. But what about the large number of flops where we have flopped ace-high only? On these flops we have a substantial amount of equity - far too much to simply give up - but we cannot play in a manner that is terribly favorable to our implied odds. Our opponent will have flopped a pair $^1/_3$ of the time and some type of thaw worth committing with an additional $^1/_{6^{th}}$ of the time. Hence, our opponent will be able to commit accurately with liis best 50% of hands. In this sense, he has better clairvoyance than we do; and he has effective position. Since he can automatically check, he effectively forces us to act first on the flop. Any strategy we employ in this situation has its drawbacks.

Checking behind risks giving a free card that beats us when we have the best hand, and exposes us to being bluffed out on the turn. To balance that, we get another card to try to make a pair as well, while taking back effective position.

Betting a small amount and folding to a check-raise all-in allows our opponent to steal the pot very frequently with hands that we beat; he can effectively turn his 40 or 50% equity in the pot into the whole pot with draws.

Betting a small amount and calling a check-raise does better against draws, but loses equity against made hands. By playing this strategy, we effectively allow our opponent to be selective about the hands with which he will commit. The same situation occurs if we simply jam the flop behind; we allow our opponent to be selective in calling our jam and allow him to identify the strength of his hand before doing this.

All these difficulties, however, are obviated by jamming preflop. At some stack size, however, it's clear that we expose too much (or alternatively, jam with too few hands) by jamming preflop. Hence, we must strike a balance between jamming outright and raising to a lesser amount. We can iterate the above analysis for a long time, looking for breakeven points. In fact, if we could estimate postflop equity very- accurately, we could likely arrive at a stack size for every position.

But as a practical matter, we advocate jamming with all hands (in lieu of raising to a lesser amount) for stack sizes of approximately 6 times the pre-raise pot from the front and 7 times the pot from near the button.

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Play on the Flop

After preflop play has concluded and the flop has been dealt, a great deal of information about the eventual values of the hands has been revealed. Using this information to play accurately on the flop by assessing the shape of the distributions accurately is a critical skill in playing no-limit holdem well.

There are two major elements we focus on in examining the effect of the flop on future play. The first is who ***the previous aggressor*** was; that is, who was the player who last bet or raised preflop? This player will generally (in an environment of near-optimal play) possess a stronger distribution of hands than his opponent. Also included in his distribution will be weaker hands that balance his stronger hands. The second is the ***texture*** of the flop, a term used loosely to describe the approximate likelihood that either player's hand contains draws, and the strength of those draws.

For example, aboard like 9♦ 8♦ 7♣ contains a large number of potential draws. The shape of distributions (no matter how carefully balanced preflop) is altered by the nature of the board. We call boards like this one either ***drawish*** or ***dynamic.*** By contrast, a board like K♠ 8♣ *2*♦ contains very few draws; in fact, most of the draws on this board will be five or six-outers to pairs or two pair. This type of board is more ***static.*** The extreme example of a static board is a board such as A♥ A♣ A♦, where there are just a few pair draws and few classes of hands overall. The ranks and texture of the flop affect the play of the distributions in ways that are extremely significant.

One aspect that is highlighted by the idea of the previous aggressor is that the distributions in real poker on the flop (or after some action has taken place) are nearly always asymmetric -that is, in contrast to many of our toy games, the players hold different sets of hands. Consider the situation where an early position raiser confronts a caller in the big blind. The blind will hold a significantly weaker distribution on the flop than the raiser, because he will have called with a much looser set of hands because of his reduced price for entering the pot. Even in situations where the stacks are deep and the raiser has a less tight set of hands, the blind's distribution will still be much weaker.

In fact, the raiser's range in this position is so strong that he likely has an autobet on most flops: that is, he should bet with all his hands if checked to. In this case, the blind should check all his hands (we can show, although we omit the proof here, that if the raiser has an autobet, then it must be at least co-optimal to check as the first player). We can see that this leads to a mode of play that gives deference to the previous aggressor (the raiser). It may be protested that the raiser can exploit this strategy of autochecking by checking behind, but we believe that the raiser will have higher equity utilizing an autobetting strategy than he would attempting to exploit by checking behind. This is because his distribution is so strong that he wants to either extract value or claim the pot; this situation is loosely analogous to the situation where a player has few bluffs, and so his opponent can reasonably fold a large portion of his hands (more than a) because he need not fear being exploited.

As the blind here, in addition to our weak hands, we will additionally have hands with value that we want to continue playing, such as pairs, two pair; sets, flush draws. straight draws, and so on. After the early raiser autobets, we must construct a strategy that maximizes value as well as being balanced. To do this, we must identify our strategic options and the hands that best fit those strategic options. After checking and having our opponent bet we can raise, call, or fold. It's clear from preceding material (and intuitively as well, we hope) that we will fold our worst hands. Additionally, we have the options of calling or raising. Assume the stacks here are quite deep.

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A thorough, treatment of these options for all different stack sizes, raiser positions, and so on is beyond the scope of this chapter. However, we can identify some principles that may help to guide us in identifying a balanced strategy. We can generally say the following:

Both we and our opponent will take actions in "action regions" that will have threshold hands. (These thresholds may more formally be sets of hands due to card removal effects, but we can still visualize in this way.) In constructing a balanced strategy, our parameterization of regions must be such that we can create the proper indifferences between actions at the threshold points so that he cannot exploit us by moving his thresholds.

This might not be that easy to see, but suppose that we consider a check-raise. Our opponent has (for most stack sizes) three reasonable options. The first is to fold to our check-raise. The second is to call, and the third is to jam (or reraise). For each of these regions, there will be thresholds - in fact, because his non-folding actions will contain semi-bluffs to balance them, there will be multiple thresholds. Additionally, because this is not a static game, hands have different values - a draw value (which is the distribution of the hand's strength on the river) and a current value (which is the distribution of the hand's strength right now).

When we raise, then, we want to raise a mixture of hands such that our opponent cannot exploit us by moving his thresholds. Suppose that player Y has a threshold between y_1 and y_2. Then his opponent must have hands in different regions such that the threshold hand is mdifferent to playing y_2 or y_1. This is a little difficult to ensure in practice, but we can effectively substitute the idea that if Y attempts to exploit him by expanding any one of the regions and playing that option more often, the opponent must have hands such that one or more regions of his hands benefit enough from that movement to offset or more than offset the loss to the other regions.

Suppose we divide all the blind's hands into the following categories:

1) Strong hands (i.e., sets, two pair, straights, and the like).
2) Medium strength made hands with weak draws (one pair hands).
3) Weak made hands with very strong draws (flush draws with additional value, such as a gutshot or pair).
4) Weak made hands with medium draws (flush draws or straight draws).
5) Weak hands without at least medium draws.

In truth, the only way to assess the relative weights of all these possibilities would be to calculate the optimal strategy' for a game with preflop distributions specified - even this is beyond current computing power. However, we can use the above to understand what a reasonable and balanced mixture of check-raising hands might be.

Hands that benefit the most from the opponent folding are draws, particularly weak ones. In fact, the weaker the draw, the more value is gained when the opponent folds. This of course must be balanced against the additional lost value against the opponent's stronger hands and even semibluffs.

If the opponent calls with more strong hands (that is, reraises less and calls more), we lose value with strong hands, as well as losing value with medium strength hands. Draws, however, benefit because they are not forced to put in additional money or fold to a reraise.

If the opponent calls with more weak hands (that is, folds less and calls more), we benefit with strong hands and medium strength hands, while losing value with weak draws.

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If the opponent reraises more often, strong hands benefit, while draws are punished - except for the special case of very strong draws. These hands can react to a reraise by pushing all-in, as we saw in Part H.

What we seek to do in constructing a balanced strategy here is to identify the hands that have high value in one action, balance those hands, and also ensure that the remaining hands (which will play the other branch of check-calling} are also balanced.

Very weak draws have high value in check-raising, because they gain so much from the opponent folding. La fact, it is not unusual for many very weak draws to be unplayable if there was no raise option, but to have positive expectation from raising. Likewise, check-raising with a sizable portion of strong hands is also indicated. This is in keeping with the idea of correlating the size of the pot to the strength of your hand. The third type of hand, which (along with strong hands) helps to defend against looser reraising , is very strong draws, such as open-ended straight flush draws and gutshot straight flush draws. These hands have a very effective response to a reraise - to shove all-in. Recall that we discussed these types of hands in Part II. They benefit from getting all the money in, even against strong opposing hands. The worst outcome for them is to get some of the money in and then have to play more streets with a short stack relative to the pot size. This is because their value degrades on each street that they do not hit. In addition, when they do hit, they are less likely to be paid off. Likewise, since we can play these high-value draws in this way, we can play some very strong hands in this manner - this picks up value if our opponent calls looser (such as with one-pair hands) expecting us to be unbalanced toward playing strong draws.

If we construct a strategy of check-raising with some very weak draws (with the intention of giving up to a raise or a call if we miss on the turn), with some very strong hands (we will need some strong hands to balance our calling distribution), and very strong draws, we will include the hands that do the best by check-raising and will have achieved balance. This leaves hands of medium strength, some strong hands, and weaker draws (such as ace-high flush draws) with which we can call. This second set of hands is also balanced because it creates a situation where after the turn card, the distribution of hands will be such that the opponent cannot exploit by betting more or less often. This is because the combination of strong hands from the original distribution and hands that have become strong from the draws will be sufficient to prevent this type of exploitation.

That's a lot of ideas and things to think about, and all we've considered here is one decision point, in a fairly simple situation where one player is autobetting. Additionally, we neglected to specify bet/raise amounts, nor have we considered the texture of the flop, and so on. Playing strong exploitive poker is hard; it requires that you gather information in situations where that information is highly ambiguous and often misleading, calculate the inference from that information, and not expose yourself to counter-exploitation too much. But playing near-optimal poker is hard as well. Considering situations rather than hands and ensuring that one's play is balanced among the various alternatives is very difficult.

We next consider some ideas related to bet sizing. The texture of the flop is extremely important in the approach we take to bet sizing because of how it impacts the shape of the distributions. The presence of two suited cards on the board adds a prominent set of hands (flush draws) into the possible hands for both players. Since the draw is open, this will affect the play significantly. Consider first, however, a fairly static board, such as KK4, where we have been the preflop aggressor with a moderately strong distribution. On this type of flop, our tendency is to bet approximately the three-street geometric growth of pot, assuming that the stacks are moderately sized (such that geometric growth is between $^1/_3$ of the pot and a pot-sized bet). This betting

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scheme gives us a fairly good balance between putting chips at risk when the opponent is strong and extracting value from his weaker draws. Since the quality of the draws on this board is very poor, we need not bet a lot in order to make the borderline draws indifferent.

When the stacks are larger, we often restrict ourselves to betting less than the pot; instead, we might use a four-street geometric growth scheme, allowing for a potential raise from our opponent to take up one of the streets. On static boards, we never overbet the pot, preferring to take advantage of the static nature of the board to provide value to our distribution.

When the board contains two flush cards or connected cards, however, the situation is different. First of all, a substantial fraction of the hands our opponent will continue with will have no value on the river. In order to make us indifferent to "bluffing" the opponent need only consider his hands that beat a bluff. What happens here is that on the river, the opponent with the weaker distribution, because of his knowledge of whether he held a wreaker draw or a stronger one, has an advantage in the betting on the river. He is effectively playing with more clairvoyance than we are on the river - and when the flush misses, he need not pay us off with those draws at all. We will refer in general to "the obvious draw", which is usually the flush draw on a board where such is possible, or a straight draw on a board like J♣ T♥ 4♠.

Because of this enhanced clairvoyance effect, we are more inclined to bet larger amounts on earlier streets, offsetting the ex-showdown disadvantage we will have on later streets. On boards that have only a moderate amount of draws, such as K♠ 7♥ 2♥, we might simply increase our bet amount by perhaps half of the three-street geometric growth of pot. Hence, if we would bet $^1/_2$ of the pot geometrically, we might bet $^3/_4$ of the pot on a board like this one. When the board is extremely draw-oriented, such as the board discussed briefly earlier (9♠ 8♠ 7♦), we will often make our bet amount the two-street geometric growth of pot, attempting to maximize against our opponent's draw-heavy distribution. And when the board has three flush cards, the distributions are even more impacted by the flop; here we often use the two-street geometric growth of pot as well.

Each of these bet amounts requires adjustments in the set of hands we play. If we are betting a larger amount, we should do so with fewer hands in order to prevent exploitation. This requires that we effectively balance our play by checking both hands with little value and hands with a lot of value. Take, for example, a three-flush flop such as K♣ 7♣ 4♣. On this type of flop, it is likely advisable (assuming that our bet amount on this flop would be approximately the pot) to check behind hands such as AQ with no club and the like. To balance this, then, we will need to check behind some hands with strong showdown value.

The danger of checking behind is that we give weaker hands a free card to beat us. Here the free card danger is exaggerated by the nature of the flop - giving a free card to a lone club can be disastrous. In trying to balance our play here, we look for hands that are not terribly hurt by giving a free card to a lone club, but have strong showdown value if blanks hit. Hands such as A♣ Ax or A♣ Kx fit this description perfectly; in a sense they are likely the best made hand and the best thaw. By checking behind on the flop, we in a sense induce a bluffer semi-bluff on the turn. It is extremely important that we gain value with some of our hands to offset this.

Play on the Turn

We next consider play on the turn. This section will be substantially shorter than the section on flop play, mostly' because many of the concepts that apply on the turn are the same as those that were discussed under flop play. The importance of balancing our play between the various strategic options, distributing our strong hands across our strategic options so that we threaten to have a strong hand from any action sequence, and so on are all just as important in turn play as

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on the flop. The two areas in which turn play differs substantially from flop play are in the realization of draw equity and the complement of that situation, when a blank has come off on the turn.

Consider first the situation where no obvious draw has come in. The hand values of "made hands" remain more or less the same, with the draws that missed being downgraded substantially because they generally have just around 20% chance of coming in. Obviously, the rank of the card that hit on the turn can change the value of some medium- strength hands, either by making two-pair or by making a higher one-pair. Ordinarily, the stronger distribution (from the previous street) has the advantage in this situation, as it is often less likely to have the draw. Typically, our continuation betting strategy is structured as follows:

Suppose that on the flop, we had bet a set of hands. Of these hands, we will now check with following classes of hands:

- A few very strong hands.
- Hands that were of moderate strength on the flop, but unproved on the turn (for example, hands which hit two-pair).
- Some hands of moderate strength that will now check-call. (This group often includes pairs that arc not top pair or overpairs or the like.)
- Some weaker hands that will now check-fold. (Often weak draws are in this category, or hands with just one overcard or the like.)

The hands that we continue to bet are:

- The other very strong hands.
- Hands of decent value, such as top pair or two-pair (with the intention of calling a raise).
- Stronger semi-bluffs (with the intention of calling a raise to balance the above).
- Some weaker semibluffs (that will fold to a raise).

The selection of hands here is not set in absolutes - obviously the nature of the board, the ranks of cards on it, and so on; each have an important effect on the selection of particular hands. What's important is that for each action we might take, we must select a distribution of hands such that our play cannot be exploited by an opponent. We know of many players who are extremely vulnerable to exploitation in a sequence where they raise preflop and get called behind. The flop comes down and they bet and get called, and they then check and fold to a bet from behind far too often relative to their checking frequency. They could make themselves less exploitable by checking more strong hands and hands that are able to call a bet on the turn.

When the draw does come in, many of the same principles apply. However, because many the of the semi-bluffs have now been promoted to strong hands, many hands which would have been check-calls arc demoted to weaker bets, while hands that would have been weaker bets might be demoted to check-folds. Likewise, when the draw comes in, often an additional draw is created. Suppose that the card that made the draw on the turn was a flush card. Then the bare ace of the Bush draw is promoted to the nut draw against made flushes or other made hands.

In optimal strategy, having the backdoor nut flush draw on the flop (the bare ace of the two-flush on the board) is of real importance. Holding that card has two important effects. First of all, it creates the potential for a runner-runner nut draw, which is of course important for equity. Additionally it decreases the chance that the opponent is holding a strong flush draw when he pushes all-in on the flop. Suppose that you hold A♣ K♦ on a flop of K♣ 7♣ 2♥, and the sequence evolves such that your opponent shoves all-in for a large raise on the flop. If you held the Ah,

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your opponent's distribution might contain some Ac Xc flush draws, expecting to often have 12 outs when called (9 flush cards and 3 aces). However, since you hold the A♣, you might give a little more weight to folding because the opponent's distribution contains none of these hands although he does not know this.

Card removal effects like this additionally have profound implications for river play, the topic we will consider in the next section.

Play on the River

The river is the simplest street to play on; it is also the street where the pot is largest and making mistakes has the highest cost. The river is the primary street where deficiencies of previous hand distribution selections turns into actual lost profit, though in the way that most players typically think of poker, this is an invisible effect. It is not too much of an exaggeration to say that the goal of a poker hand is to arrive at the river with a distribution that contains a mixture of strong, medium strength hands that did not reach their full potential due to the cards that fell, and weak hands and a pot size correlated to the strength of that distribution.

Play on the river, then, is quite comparable to the [0,1] game with asymmetric distributions, except for the fact that the hand ranks are discrete instead of continuous, and there are card removal effects. In this light, we find ourselves often following through on value betting the geometric growth of pot on the river with hands that are appropriately strong.

However, taking a lesson from the no-limit AKQ game, we additionally employ "preemptive bets" on the river. These are small bets that we make with a mixed distribution of hands that have some showdown value, but don't want to call a large bluff on the river, and hands that are very strong. By doing this, we force the opponent to either raise us with their bluffs, in which case we win more money with our strong hands, or give up on those hands, in which case we gain from not losing the pot when they would have bluffed. Against that we lose additional money when the opponent raises, as we typically have to fold against a raise on the river. These preemptive bets are often as little as $^1/_{10}$ to $^1/_6$ of the pot.

The most common scenario in which we will make these types of bets is when the obvious draw has come in but our distribution contains a significant number of hands that were made by the flush card as well. In these cases, our distribution still contains a number of hands that are of mediocre strength and don't want to call a big bluff on the end, but still have value. Thus we make preemptive bets with a mixture of flushes and pairs (usually), along with standard value bets with some flushes and other hands.

Card removal also plays a significant role in deciding which hands to bluff. In general, when selecting from among the hands that have little enough showdown value to bluff, we should select hands which make it more likely that our opponent does not have a calling hand. In the same way, we should pay attention to our cards when deciding whether to call down an opponent who might be bluffing.

Consider a seven-card stud hand where the opponent has a marked flush draw and we have aces up. On the river, the opponent is effectively clairvoyant (assuming we do not have a full house). He will bluff to make us indifferent to calling if we do not catch one of his flush cards, because that is the most likely scenario. He cannot make us indifferent to calling if we catch one of his flush cards; hence we should simply call all the time when we do catch one, and an appropriate additional fraction of the otherwise.

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Key Concepts

- Sometimes it may seem that you are being exploited on a particular street while playing near optimally. However, often this occurs because the opponent has sacrificed equity on a previous street in order to obtain the position where he can exploit on this street. For example, consider the game we studied in the multi-street section where the exposed hand needs only call $^1/_6$ of the time on the river because his opponent was forced to call a turn bet in order to see the river.
- For every action sequence, there should be corresponding balancing regions. Value, semi-bluffing (when there are cards to come) and bluffing regions should generally be seen together in a strategy.
- In every action sequence, provided there is a significant amount of action remaining, it is imperative to threaten to have a strong hand so that you cannot be exploited by the opponent wantonly overbetting the pot.
- The texture of the board affects the hand distributions profoundly, in constructing distributions for play on early streets, it's necessary to understand the expectation across all possible boards. This, of course, is impossible in practice due to computational infeasibility.

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Part IV: Risk

"My ventures are not in one bottom trusted, Nor to one place; nor is my whole estate Upon the fortune of this present year: Therefore my merchandise makes me not sad."

William Shakespeare, *"The Merchant of Venice"*

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